



Written Examination

2023-08-23

Name: _____ First Name: _____

Program of Study: _____

Matr.-No.: _____

Information:

- Write your name and matriculation number on **every sheet of paper**.
- Make sure that all **10 pages** are present.
- Duration of the exam: **90 minutes**.
- **No additional aids** (notes, calculator, etc.) are allowed.
- Use a pen with **blue or black ink**. Text written with **pencils or red/green pens will NOT be graded**.
- Write **legibly**. Unreadable answers will not be graded.
- Answer each question on the provided sheet. If you need more space, use the two blank pages at the end or ask for more paper. In these cases, please make a short note in the answer box, so we can find your answer (e.g. "see last page").
- If you have to draw something to answer a question, multiple templates are provided.
Cross out wrong answers!
- At the end of the exam you must hand in this cover sheet together with the question sheets and all additionally used paper.

With my signature I confirm that I have **read and understood** the information above.

Signature

Question:	1	2	3	4	5	6	Total	Grade
Points:	7	9	6	7	8	8	45	-
Score:								
Moderation:								

DoP
required

Question 1: Image Filtering and Pattern Recognition ($\Sigma = 7$)

- (a) What is the output of correlating the image below with the provided filter? Assume (/2) we are same-padding with zeros.

1	2	3	4
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Image

-1	1
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Filter

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- (b) Starting from an input gray-scale image I , describe how to find all **horizontal** lines (___/5) in the image using Hough Transform, explicitly stating the voting equations (you do not have to fill all lines below).

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____

Question 2: Local Features ($\Sigma = 9$)

- (a) Write a function that computes the Harris corner response on an input image with (___/5) given sigma value. You can assume that the following functions are given:

- `gauss(image, sigma)`
smooths an image with gaussian kernel
- `gauss_first_derivs(image, sigma)`
computes the first order derivatives of an image
- `gauss_second_derivs(image, sigma)`
computes the second order derivatives of an image

Use python syntax and assume numpy is imported. Loops are not allowed in this task. Small syntax errors are allowed as long as the meaning is clear.

```
1 import numpy as np
2
3 def harris_scores(image, alpha, sigma_deriv, sigma_window):
4     """
5     Args:
6         image: HxW array of an image
7         alpha: detector parameter
8         sigma_deriv: sigma for computing image derivative
9         sigma_window: sigma for the window function
10
11     Returns:
12         scores: Harris score map
13     """
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15     # START HERE
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42     return scores
```

- (b) Can the idea of Harris corner detector be adapted to identify textureless region? If (___/2) yes, explain the approach. If not, provide a justification.

- (c) You want to find correspondences between the following two images (they are related (___/2) only by translation). Would raw pixel values work well as local descriptors in cases like this? Justify your answer.



Question 3: Image Segmentation.....($\Sigma = 6$)

- (a) We want to apply MRF-based graph-cut segmentation on a rather small image with (___/4) only two pixels a and b . Given below are the unary potentials, the pairwise potential is 1. Draw the corresponding flow-graph in the box below. Remember that nodes connected to the source s after the graph cut are assigned to the foreground.

	a	b
foreground potentials:	5	0
background potentials:	1	4

- (b) We want to constrain the graph-cut segmentation such that all pixels on the image (___/2) border are always assigned to the background. How do we need to adapt a flow-graph to implement this constraint?

Question 4: Sliding-Window Object Detection ($\Sigma = 7$)

- (a) What is the difference between a classifier and a detector? (___/2)

- (b) What is motivation behind gradient-based representations like HOG and why are they well-suited for pedestrian detection? (___/2)

- (c) We want to apply a HOG-detector for car detection. Given is a dataset of 1000 cropped car images and 1000 non-car images, each of size 200×100 pixels. We first extract HOG-features, where each cell has a spatial size of 10×10 pixels and 8 orientation bins, then a linear SVM is trained. After training, we apply the HOG-detector on an image of size 2000×2000 pixels.

- i. What is the size (number of scalar values) of a HOG feature vector? (___/1)

- ii. How many learnable parameters does the SVM have? (___/1)

- iii. How many feature vectors do we get for training the SVM? (___/1)

Question 5: Convolutional Neural Networks..... ($\Sigma = 8$)

In this task, we want to develop a CNN which can colorize grayscale images. Given a grayscale image as input, the network should return an image showing the same content but in color. CIFAR-10 is used for training, which consists of RGB images of size 32×32 . We compute the grayscale image by averaging the color channels.

- (a) The first convolution layer has a kernel size of 3×3 , uses a bias, produces 100 output channels, is applied with a stride of 1 and pads with zeros with a width of 1. How many parameters does this layer have? What is the number of values in this layer's output (using a batch size of 1)?

- (b) We decide to use an encoder-decoder architecture. What options do we have for decreasing the spatial size of the feature maps in the encoder? Name two and describe them briefly.

- (c) We decide to use the L_1 loss for this task. Provide the formula for this loss when applied on a minibatch of n samples (x_i, t_i) , and explain the meaning of each variable.

Question 6: 3D Geometry ($\Sigma = 8$)

- (a) Write out the *intrinsic camera matrix* in its most general form. Explain the meaning (___/2) of each parameter.

(b) Given two projection matrices

(____/6)

$$P_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad P_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

and a pair of correspondences

$$x_1 = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \end{bmatrix}^T \quad x_2 = \begin{bmatrix} \frac{1}{3} & \frac{4}{3} \end{bmatrix}^T,$$

triangulate the location of the 3D point $X = (a, b, c)$ using the linear algebraic approach.Answer: $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (write intermediate steps in the box)

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Use these pages if you run out of space in the solution box.
Please add a reference pointing to this page from the box of the question.

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Please add a reference pointing to this page from the box of the question.